MI Live

Control a bot by imagining left or right hand movement

User guide - what each step does and how to use it

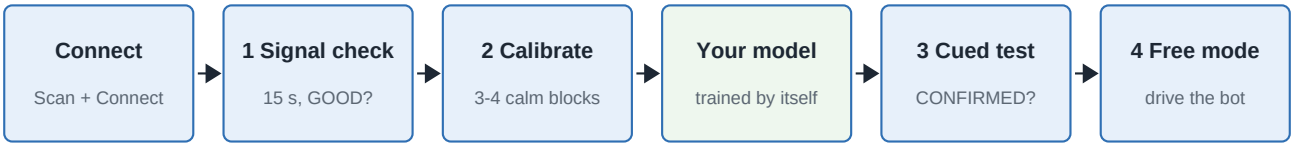
EEG motor-imagery project (D:\trainsvm, apps/mi_live)
Headset: Neiry / Capsule LEHeadPhones2 (C3, C4, A1, A2 at 250 Hz)
Version of 26 October 2026

1. What MI Live is

When you **imagine** squeezing your left or right hand, the brain rhythms over the motor area on the **opposite** side become weaker for a few seconds (mu 8-13 Hz and beta 13-30 Hz). Imagining the left hand weakens them at C4 (right side of the head); imagining the right hand weakens them at C3. This is called **ERD** (event-related desynchronisation). MI Live measures C3 and C4 with the headphones, decides 25 times a second whether the pattern looks more like LEFT or RIGHT, and turns that into a bot command: **LEFT**, **RIGHT** or **STOP**.

Two kinds of models are used. A **base model** trained on BCI Competition IV dataset 2b (9 people, recorded in Graz in 2008, downloaded) works for a new person without any training, but less well on our dry headset. Your **own model** is trained from your calibration and usually suits you better. The app tells you honestly whether a model really works for you before you rely on it.

The whole session



What you need

Item	Details
Headset	Neiry / Capsule headphones (LEHeadPhones2, shown as USAMS-YX05), charged
Laptop	Windows with Bluetooth, **running on battery** (the charger adds 50 Hz hum)
Software	C:\Python313\python.exe with numpy, scipy, scikit-learn, sounddevice (installed). The app folder is D:\trainsvm\apps\mi_live
Time	about 25-30 minutes for a full session (check, calibration, test, free mode)
Optional bot	Arduino 2-wheel bot over USB / Bluetooth serial, or any device on Wi-Fi (UDP)

Start the app

Double-click	What it opens
run.bat	the real app with the headphones
run_simulator.bat	practice without a headset: hold the Left / Right arrow keys to 'imagine'
run_demo_2b.bat	a demo with real competition EEG (person B04) played as if live - shows what a working BCI looks like (about 95 % correct)

2. Quick start (one page)

Before you sit down

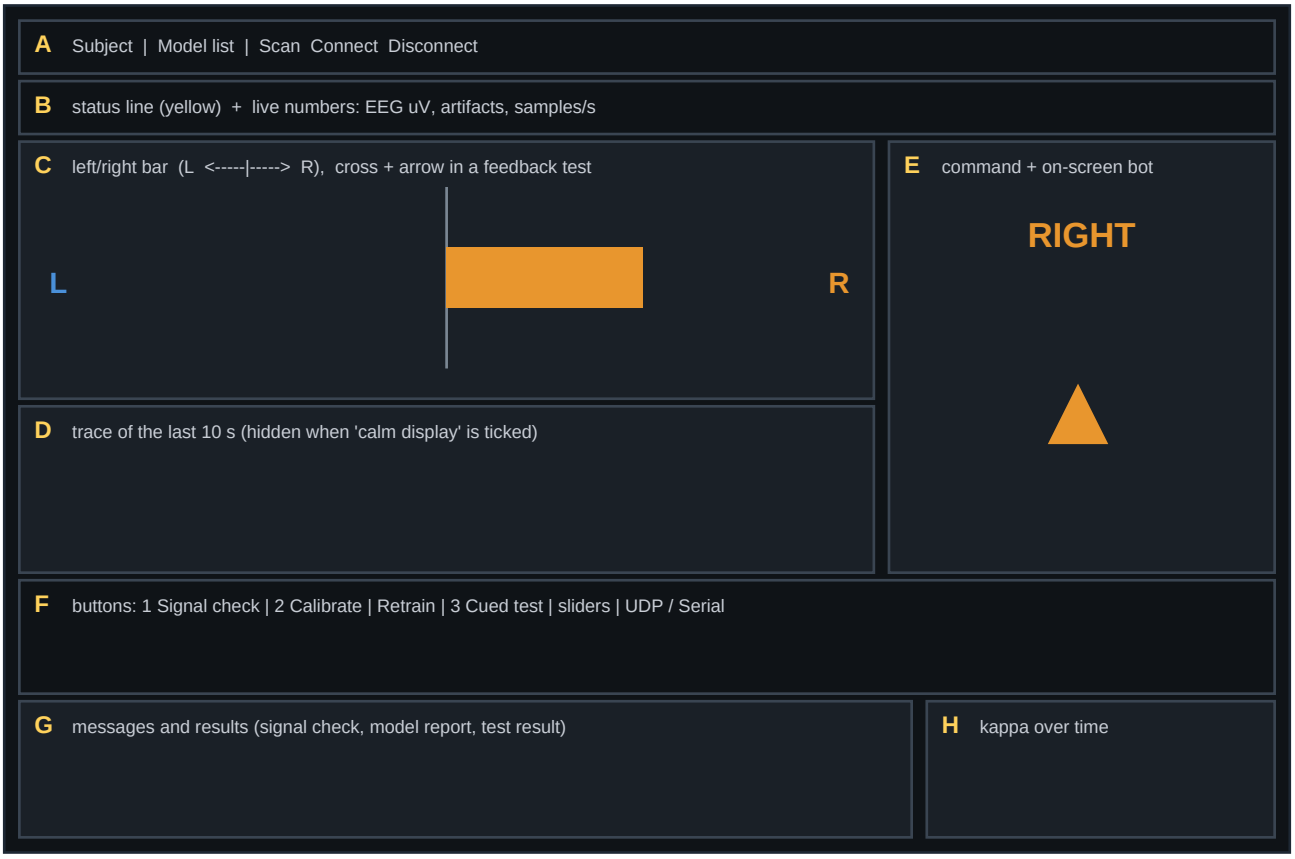
- Unplug the laptop charger. Put the laptop where you can see it without turning your head.
- Put the headphones on the right way round (L on the left ear). Press the C3 / C4 pads onto the scalp (part the hair under them) and make sure the ear clips touch skin.
- Close other programs that use the headset (mi_bci, the Capsule app). Only one program can connect.

1. **Start:** double-click **run.bat**. Type your name in **Subject** (always the same spelling).
2. **Connect:** press **Scan** (8 s), choose the headphones, press **Connect**. Wait about 10 seconds.
3. **1. Signal check:** sit still for 15 s. Repeat until it says **GOOD**; follow its advice if it says FAIR or BAD.
4. **2. Calibrate:** a black full screen opens. Press **Space**, look at the cross, and when the small arrow appears **imagine** squeezing that hand until the cross disappears. 3-4 blocks, Space between blocks. Your model is trained automatically at the end (about 1 minute).
5. **Read the model report** in the message box: PROMISING or NOT READY.
6. **3. Cued test:** same calm screen, new trials. If it says **CONFIRMED**, your model is READY.
7. **4. Free mode:** just imagine left or right. The big word and the bot follow after about 2-4 seconds.
8. **Finish:** close the window (the bot gets STOP and the headset disconnects).

Keys

Space	start the next block on the calm screen
Esc	stop the running test or calibration (what was recorded is kept)
Left / Right arrow	simulator only: pretend to imagine left / right

3. The main window



Area	What it shows / does
A	**Subject**: your name - recordings and models are kept per name. **Model**: which model decodes now (newest of yours first; the base 2b model at the top). **Scan / Connect / Disconnect** for the headphones.
B	Yellow status line: what the app is doing or what to do next. Grey line: EEG strength at C3 and C4 (uV), % of outputs flagged as artifacts, raw output d and smoothed output v, alignment state and samples per second (should be about 250).
C	The left/right bar. It grows to the left or right with the smoothed output v (from -1 = all left to +1 = all right). The dashed lines are the threshold. During a test with feedback the bar turns green when it points to the cued side and red when not.
D	The last 10 seconds of output. Hidden while 'calm display' is ticked (default), because a jittering line pulls the eyes and attention.
E	The command sent to the bot (LEFT / RIGHT / STOP / FORWARD, or TEST / CHECK / TRAINING / NO DATA) and the on-screen bot that obeys it. 'Reset bot' puts it back in the middle.
F	The step buttons and settings (see sections 4 to 8).
G	Messages: signal check numbers and advice, your model report, every test trial and the test result.
H	Kappa over time after the cue for the last test (the competition score; section 7).

4. Connect and Signal check

Connect

Scan searches Bluetooth for 8 seconds; pick the headphones and press **Connect**. The app starts the 250 Hz EEG stream. Give it about 10 s: dry electrodes need a moment to settle. If nothing is found, check that the headphones are on and charged, that Bluetooth is on, and that no other program is connected to them.

Step 1 - Signal check (15 s)

What you do: press the button and sit still with your eyes open on the screen, jaw relaxed.

What it does: it records 15 s of C3-A1 and C4-A2 and measures:

Measure	Meaning	Good value
EEG 8-30 Hz (uV)	strength of the brain rhythms the decoder uses	about 3 - 40 uV
50 Hz hum (uV)	mains electricity picked up by loose contact or the charger. Hum is much bigger than EEG and can leak into the decoder	below 10 uV (GOOD)
Peak-to-peak	big jumps from movement, jaw clenching, cable pulls	below 1500 uV
Alpha peak	the frequency of the strongest 7-14 Hz rhythm (just information)	about 8-12 Hz

Verdict	What to do
GOOD	go on to Calibrate.
FAIR	usable, but improve contact if you can and check again.
BAD	fix first: unplug the charger, press the named pad and its ear clip firmly onto skin, move away from cables and power bricks, keep still. Then check again. The app warns you if you try to calibrate on a BAD signal.

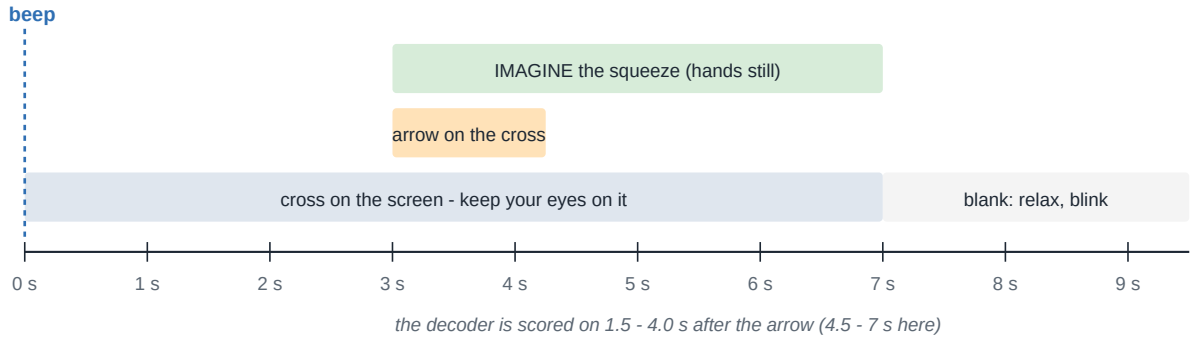
Why this matters

On the first evening the hum at C3 was 54-69 uV while the EEG was only 6-8 uV, so the decoder mostly saw electricity, not the brain. After fixing the contact and unplugging the charger it was 2-4 uV. No model can work on a BAD signal.

5. Step 2 - Calibration on the calm screen

A separate black window fills the screen. It shows only a cross and, for about one second, a small arrow on the cross. Nothing moves, there are no numbers and no feedback, so nothing distracts you. Left and right trials are mixed in random order (never more than 3 of the same in a row).

One trial



the decoder is scored on 1.5 - 4.0 s after the arrow (4.5 - 7 s here)

Time	Screen	What you do
0 s	cross + beep	look at the centre of the cross and stay still
3 s	small arrow on the cross (1.25 s)	start **imagining** squeezing the hand it points to
3 - 7 s	cross only	keep imagining a strong, slow squeeze you can feel; hands, face, eyes still
7 - 9.5 s	empty	stop, relax, blink, swallow

Blocks

Choose the number of **blocks** (default 3, recommended 3-4) and **trials per hand** per block (default 10, so 20 trials, about 3 minutes per block). Between blocks the screen says 'Block k done': rest, move, and press **Space** when ready. **Esc** stops; finished blocks are kept.

Tick **+ real-squeeze warm-up block** to start with one block where you **really** squeeze (a firm fist). Real movement gives a much stronger pattern than imagination; it is a check that the electrodes see your hands at all and on the correct side. It is not used to train the imagery model.

How to imagine well

- Imagine the **feeling** of squeezing (pressure in the fingers, tension in the forearm), not a picture of a hand.
- Use the same imagery every time; a slow, strong squeeze held for the whole 4 seconds.
- Keep your eyes on the cross. Eye movements are picked up strongly by the ear clips and can fool the decoder.
- Do not move the hands, jaw or shoulders. Blink in the empty pause, not during the trial.

What happens inside

- Each block is saved as its own recording in D:\trainsvm\data\mi_bci\<Subject>\ (eeg.csv, events.csv, outputs.csv, run.json). The older mi_bci app can use them too.
- At the start of calibration the alignment is restarted, so the first unsettled seconds after connecting are not used.
- When the last block ends, your model is trained automatically (next section).

6. Your model (trained automatically)

After calibration, MI Live trains 9 candidate models and picks the best one. It uses all of today's cued runs of this subject (tick **use all days** to include earlier days). Runs whose signal was BAD are left out and listed in the report. **Retrain my model** does the same at any time.

	2b only	2b + yours	yours only
filter-bank CSP + SVM (9 bands, 4-40 Hz)	x	x	x
CSP + SVM (8-30 Hz)	x	x	x
log power + SVM (8-30 Hz)	x	x	x

'2b only' does not use your data; '2b + yours' gives your trials together the same weight as all 2b trials; 'yours only' ignores 2b.

How it is scored: leave-one-run-out

Each candidate is trained on all your runs except one, and tested on the run it never saw; this is repeated for every run. The score is the share of trials decided correctly (decision = the average sign of the outputs 1.5-4.0 s after the arrow). Because 9 candidates are tried, the best one could look good by luck, so its p-value is multiplied by 9.

Reading the report

Line	Meaning
left out: ...	runs not used because the signal was BAD
left/right pattern (ERD) mu / beta	is the expected opposite-side drop present in your data? PRESENT ($p < 0.05$) is a very good sign
list of 9 candidates	accuracy on runs never seen, with the binomial p (chance = 50 %)
CHOSEN	the best candidate; it is fitted on all your runs and used from now on
VERDICT	**PROMISING**: above chance after the x9 correction (and at least 60 %). **NOT READY**: not distinguishable from guessing yet

PROMISING is not yet READY

A model only becomes **READY** when a cued test on **new** trials is also above chance. On the first evening a model scored 65 % in leave-one-run-out but only 4/20 on the next run, so the extra confirmation step was added.

Your models are saved in data\mi_bci\<Subject>\decoders_live\ and appear in the Model list as 'YOURS date: name (score, verdict)'. The newest is selected automatically after training.

7. Step 3 - Cued test

The honest check. It uses the same calm full screen and the same trial timing as the calibration, with new random trials, and the model currently selected. Set **trials/hand** (10 is a good minimum; more trials give a surer answer). By default there is no feedback; tick **show feedback bar** for a Graz-style test in the main window where the bar shows the decoder during each trial.

The result in the message box

Line	Meaning
Trial k: cued LEFT -> decoded LEFT OK	each trial as it finishes (feedback test only)
Accuracy, binomial p	share of trials correct, and the chance of getting at least that many by guessing. p < 0.05 = ABOVE CHANCE
Confusion matrix	rows = what was cued, columns = what was decoded. A good model has big numbers on the diagonal. All in one column = the model is biased to one side
Competition metric: max kappa	how BCI Competition IV scored: kappa at each moment after the cue, best moment reported. 0 = chance, 1 = perfect. Compare it with the red 'chance' line; with few trials, chance can reach 0.5-0.6
CONFIRMED / NOT CONFIRMED	only for your own models: CONFIRMED (p < 0.05) marks the model READY for free mode

How many correct is 'above chance'?

Trials	20	30	40	60	80
needed for p < 0.05	15 (75 %)	20 (67 %)	26 (65 %)	37 (62 %)	48 (60 %)

A test is saved like a calibration block, so it also becomes training data for the next model.

8. Step 4 - Free mode

Whenever no check, calibration, test or training is running, MI Live is in free mode: just imagine left or right and watch the big word and the bot. A change of imagery shows up after about 2-4 seconds (2 s of EEG per output plus 2 s of smoothing).

Setting	Default	Effect
Threshold |v|	0.5	how one-sided the last 2 s must be (0.5 = 75 % of outputs agree). Higher: fewer but surer commands
Hold (s)	0.5	how long it must stay past the threshold before the command changes. Higher: slower but steadier
When undecided	STOP	STOP: the bot stands still between commands. FORWARD: it drives straight
remove bias	on	removes a constant lean to one side (the running mean of the output)
keep adapting alignment	on	keeps following slow changes of today's signal. Untick to freeze it as it is
calm display	on	hides the jittery trace and pointer; the bar, word and bot stay

9. Connecting a real bot

The command can leave the laptop in two ways at the same time as the on-screen bot. A message is sent when the command changes and again every 0.25 s, so a bot can stop by itself if messages stop.

Way	How	What is sent
UDP (Wi-Fi)	tick 'Send to bot by UDP' and type the bot's address, e.g. 192.168.1.50:5005	one JSON line: {"cmd": "LEFT", "v": -0.72, "d": -1.31, "t": ...}
Serial (USB / Bluetooth HC-05)	type the port (e.g. COM5), press Open	one letter + newline: L, R, S (stop), F (forward)

Ready-made examples

- **bot_examples\arduino_bot\arduino_bot.ino**: a 2-wheel Arduino bot with an L298N motor driver (left motor pins 5/6, right motor 9/10, 115200 baud). L and R spin on the spot, F drives, S stops; it stops by itself after 1 s without messages.
- **tools\udp_listen.py**: prints the UDP commands (run it, tick UDP with 127.0.0.1:5005). Use it as a starting point for a Raspberry Pi bot.

Safety rules built in

- STOP during the signal check, calibration, cued test and training.
- STOP when the signal is an artifact (movement, jaw) for more than 0.5 s.
- STOP and 'NO DATA' when no EEG arrived for 1 s (Bluetooth drop).
- STOP when the window is closed. Keep the real bot slow and on the floor.

10. How it works inside

one new output every 40 ms (25 per second); everything uses only past samples

Step	What and why
Montage	C3 minus left ear (A1), C4 minus right ear (A2): removes what both share, e.g. large electrode offsets
50 Hz notch	removes mains hum; the 2b training data was recorded with the same notch
Filter bank	9 bands of 4 Hz from 4 to 40 Hz (the 2008 competition winner's idea); causal filters (past samples only)
Covariance	for each band, how strong C3 and C4 are and how they move together over the last 2 s
Alignment	'Euclidean Alignment': every covariance is re-centred on the mean of today's data, so differences between days, people and headsets matter less. Uses no labels
CSP + SVM	CSP finds the C3/C4 mix whose power differs most between left and right; a linear SVM turns the 18 numbers into one output d (negative = left)
Bias, smoothing	the running mean of d is removed; v = average sign of d over the last 2 s
Command	LEFT if v <= -threshold for 'hold' seconds, RIGHT if v >= +threshold, else STOP

11. How good is it?

On the BCI IV 2b data, each person decoded by a base model that never saw them, exactly as live: **77.7 %** of trials correct on average (57 % to 95 % per person), competition kappa 0.60. On our dry headphones the first sessions (26 October 2026) were not yet above chance: a stable left/right pattern was not found. That is common at the start; imagery usually improves with practice, a GOOD signal and more calibration blocks. If it stays at chance after about three sessions, use **real hand squeezes** for control: the pattern is much stronger.

12. Troubleshooting

Problem	What to do
Scan finds nothing	close mi_bci and the Capsule app; switch the headphones off and on; Bluetooth on; scan again
Signal check BAD: hum	unplug the charger; press the named pad and its ear clip onto skin; move away from cables
Signal check BAD: no EEG / huge EEG	a pad is not touching the scalp, or you are tense; adjust the headphones, relax
'NO DATA' / samples/s below 225	Bluetooth is losing data: sit closer, reconnect
Many artifacts (> 20 %)	relax jaw and forehead, keep still, blink only in the pauses
Calibration screen text in a corner	fixed on 26 Oct 2026; if seen, move the window or press Esc and start again
Model NOT READY	keep the signal GOOD; record 1-2 more blocks; check the real-squeeze block
Test NOT CONFIRMED	do not rely on free mode yet; calibrate more blocks, retrain, test again
Bar stuck on one side	keep 'remove bias' and 'keep adapting' ticked; give it 30 s, or calibrate
An error message	see D:\trainsvm\apps\mi_live_out\error_log.txt

13. Files

Where	What
apps\mi_live\run*.bat, live.py	start the app (headset, simulator, 2b demo)
data\mi_bci\<Subject>\<date_time>_screening\	each calibration block or test: eeg.csv, events.csv, outputs.csv, run.json
data\mi_bci\<Subject>\decoders_live\	your models (live_*.json) and reports (report_*.json)
apps\mi_live\models\base_2b_live.json	the base model trained on BCI IV 2b
apps\mi_live\tools\	session_summary.py (a day's runs and results), train_personal_cli.py, check_test_run.py, eye_check.py, swap_check.py, udp_listen.py, and checks
apps\mi_live\README.md	technical notes and verification results

Useful command: `C:\Python313\python.exe tools\session_summary.py <Subject>` (run in apps\mi_live) lists today's runs with signal quality, test results and the model report.

14. Words used

Word	Meaning
Motor imagery (MI)	imagining a movement without doing it
ERD	the drop of mu/beta power over the motor area opposite the (imagined) moving hand
C3, C4, A1, A2	electrode places: left and right motor area, left and right ear
Artifact	a signal that is not brain: blinks, eye movements, jaw, movement, cable pulls
Cued trial	a trial where the screen tells you which hand (the label is known)
Leave-one-run-out	test each model on a recording it was not trained on
Binomial p	the chance of scoring at least this well by pure guessing
Kappa	accuracy corrected for chance: 0 = guessing, 1 = perfect